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END SEMESTER EXAMINATION - 2025

Name of the Course: B.Tech ECE

Semester: Vth

Name of the Paper: Microcontroller and Embedded Systems

Paper Code: **TEC-503**

Time: 3Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1	(20marks)	
(a)	Explain the features of 8051 microcontroller. Discuss the Harvard architecture and Von Neumann architecture. What are their advantages and disadvantages?	CO1
(b)	What is the use of SCON, TCON and TMOD registers in 8051? Explain with format.	
(c)	Explain the concept of bitwise operations with few instructions. Write an assembly program to toggle the 3rd and 5 th bit of a given 8-bit number stored in the memory location 20.	
Q2	(20 marks)	
(a)	Write a program for 8051 to find average of 5 numbers stored from location 20 store result at 50.	CO2
(b)	Differentiate a) MOV, MOVX and MOVC b) RRC and RR c) CLR and CPL	
(c)	How does 8051 differentiate between external and internal memory? Explain different addressing modes of 8051 with example of each.	
Q3	(20 marks)	
(a)	Write a program in 8051 to generate a square wave of 20KHz frequency at pin P1.2 using timer 2. Assume XTAL= 12MHz	CO3
(b)	What is the priority of interrupts in the 8051? Write a program to handle an external interrupt (INT1) which toggles an LED connected to P1.2.	
(c)	Explain any four logical instructions in 8051.	
Q4	(20 marks)	
(a)	Explain the important features of ATMEGA328P. Write a program to subtract two 16bit numbers with ATMEGA328P.	CO4
(b)	Explain the sketch of Arduino program also explain basic predefined functions void setup, void loop, delay and pinMode.	
(c)	Describe the concept of PWM (Pulse Width Modulation) and explain how to use PWM for controlling the brightness of an LED in Arduino.	
Q5	(20 marks)	
(a)	Interface an LCD display with 8051 to display message "INDIA" and scroll it towards left.	CO5 and CO6
(b)	Interface ADC with 8051 and transmit the converted data serially with baud rate 4800bps by serial port.	
(c)	Write an Arduino program to interface a DC motor using a motor driver (L298). The program should make the motor rotate in both clockwise and anti-clockwise directions.	

